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CUBICLE 7:

NAMES

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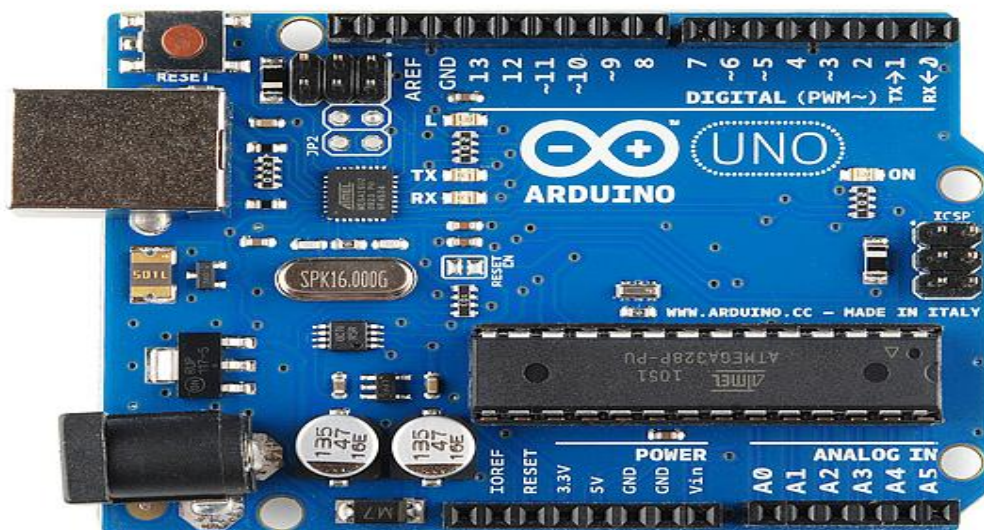
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ARDUINO BASIC:



Arduino is an open-source electronics platform based on easy-to-use hardware and software. It is designed to make working with electronics and programming more accessible for beginners, students, and professionals.

At the heart of an Arduino board is a microcontroller, which can be programmed to sense the environment and control outputs like LEDs, motors, or sensors based on input data.

PIN DESCRIPTION:

Pin Name	Type	Description
Digital Pins (0–13)	Digital I/O	Can be used as input or output for digital signals (HIGH/LOW).
Pin 0 (RX)	Digital I/O	Used for Serial Receive (UART communication).
Pin 1 (TX)	Digital I/O	Used for Serial Transmit (UART communication).
Analog Pins (A0–A5)	Analog Input	Reads analog signals (0–5V) ; can also be used as digital I/O.
PWM Pins (3, 5, 6, 9, 10, 11)	Digital PWM Output	Used for Pulse Width Modulation (e.g., dimming LEDs, controlling motors).
GND	Power	Ground pin. Common ground for all components.
5V	Power	Provides regulated 5V output to power sensors/modules.
3.3V	Power	Provides 3.3V output (less current capacity).
Vin	Power	Input voltage pin (used to power board from external supply, 7–12V).
AREF	Reference Voltage	Used to set an external analog reference voltage for ADC.
RESET	Control	Resets the microcontroller when pressed or pulled LOW.